

Types of Cryptanalytic Attacks

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- Cryptanalysis aims to comprise one or more of

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- **Attacks characterized by type of information used**

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- Cryptanalysis aims to comprise one or more of
 - Confidentiality
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 - Authentication
- Attacks characterized by type of information used
- Side Channel Attacks / Social Engineering

Mathematical attacks target the underlying algorithm

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- Ciphertext-only attack

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- Ciphertext-only attack
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- Ciphertext-only attack
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- Ciphertext-only attack
- Known-plaintext attack
- Chosen-plaintext attack
- Adaptive-plaintext attack
- Chosen/adaptive ciphertext attack
- Related-key attack

Side channel attacks target weaknesses in the implementation

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- Power usage

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- Power usage
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- Power usage
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- Differential failure

**Social engineering attacks the user to trick
or coerce them into revealing the key**

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- Rubber hose cryptanalysis

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- Rubber hose cryptanalysis
- Careless user behavior

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- Rubber hose cryptanalysis
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- Rubber hose cryptanalysis
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- Rubber hose cryptanalysis
- Careless user behavior
- Weak/default passwords
- Pretending to have legitimate access
- Naive trust

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- Rubber hose cryptanalysis
- Careless user behavior
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- Freeloaders

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- Rubber hose cryptanalysis
- Careless user behavior
- Weak/default passwords
- Pretending to have legitimate access
- Naive trust
- Freeloaders
- Willing insiders

In Review – Plenty of attacks to choose from, and only one needs to succeed

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In Review – Plenty of attacks to choose from, and only one needs to succeed

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- **Technical attacks target the algorithms**

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- **Technical attacks target the algorithms**
- **Side channel attacks target the implementation**

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- **Technical attacks target the algorithms**
- **Side channel attacks target the implementation**
- **Human nature is an attack all by itself**

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- **Technical attacks target the algorithms**
- **Side channel attacks target the implementation**
- **Human nature is an attack all by itself**
- **Social engineering attacks target the user**
- **Security and utility are competing goals**